Universidad de Costa Rica

Escuela de Ciencias de la Computación e Informática

CI-0146 Compilers

Deterministic Finite Automata(DFA)-Exercises

Professor:

Dr. Francisco J. Torres-Rojas

Students:

José Guerra — C33510
Mario Cordero — C22306
Daniel Gómez — C23310

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Exercise 12

Description: The set of strings over $\{a, b, c\}$ in which all the a's precede the b's, which in turn precede the c's. It is possible that there are no a's, b's, or c's.

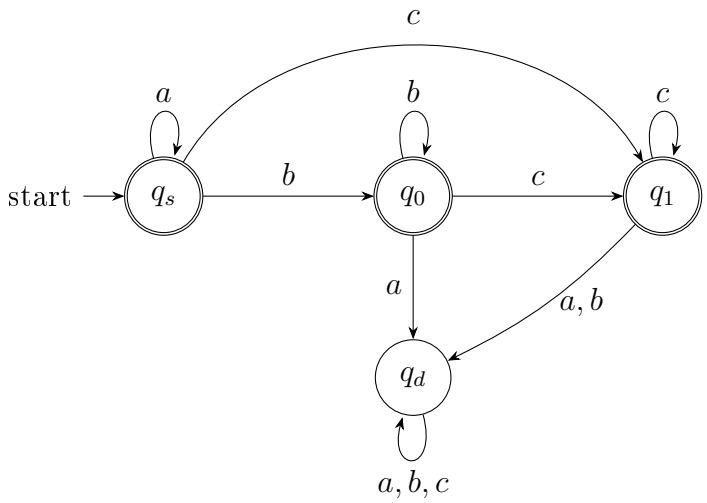


Figure 1

Exercise 13

Description: The same set as Exercise 12 without the null string.

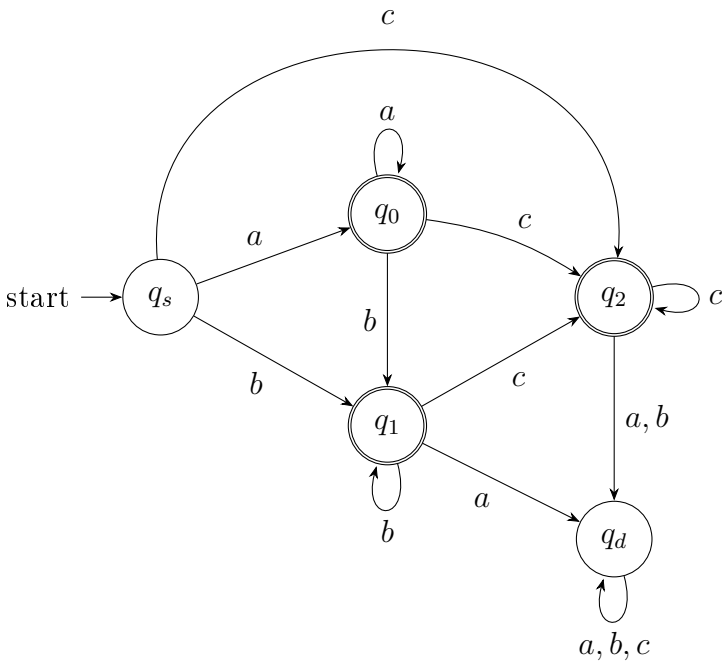


Figure 2

Exercise 14

Description: The set of strings of length two or more over $\{a, b\}$ in which all the a's precede the b's.

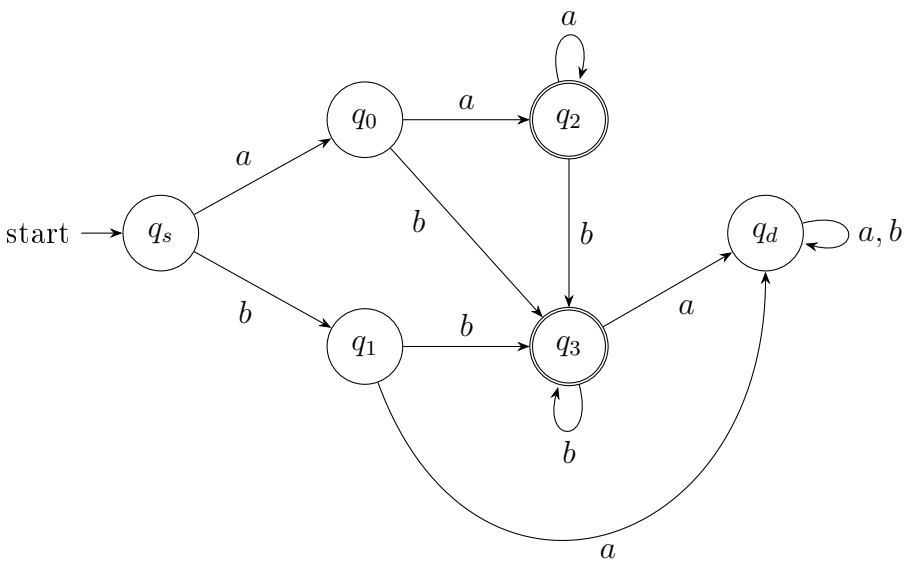


Figure 3

Exercise 15

Description: The set of strings over $\{a, b\}$ that contain the substring aa and the substring bb.

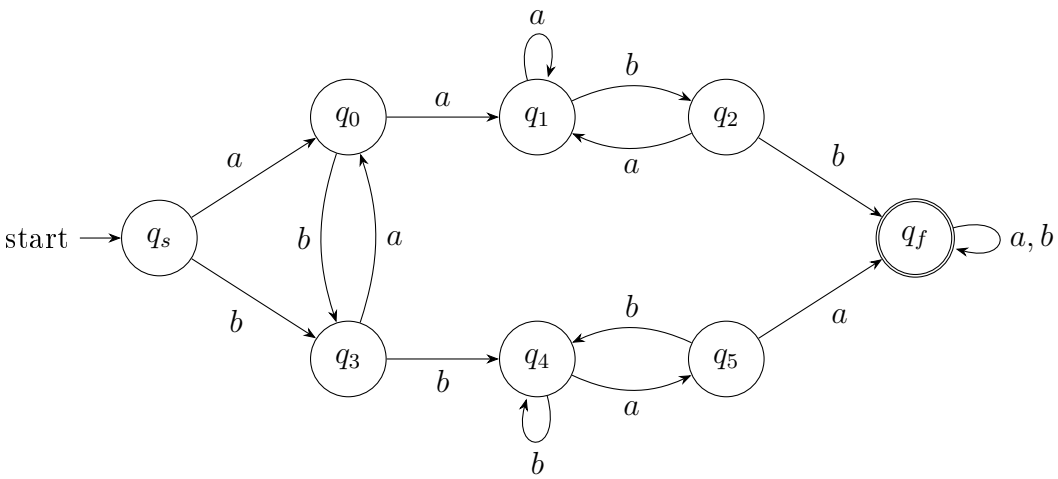


Figure 4

Exercise 16

Description: The set of strings over $\{a, b\}$ in which the substring aa occurs at least twice. Hint: Beware of the substring aaa.

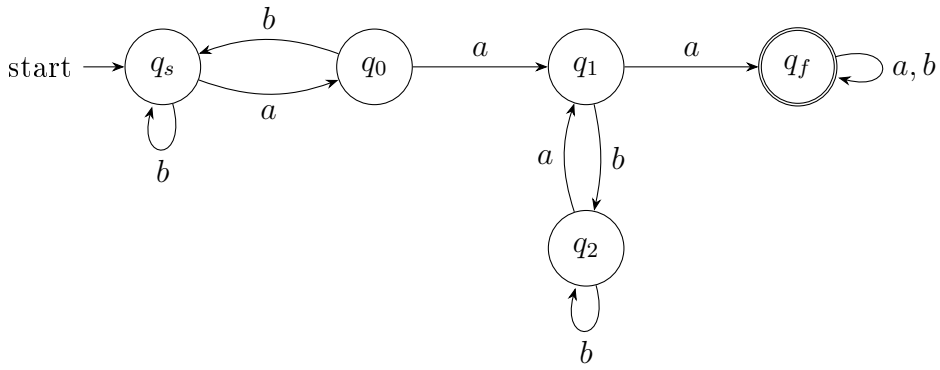


Figure 5

Exercise 17

Description: The set of strings over $\{a, b, c\}$ that do not contain the substring aa.

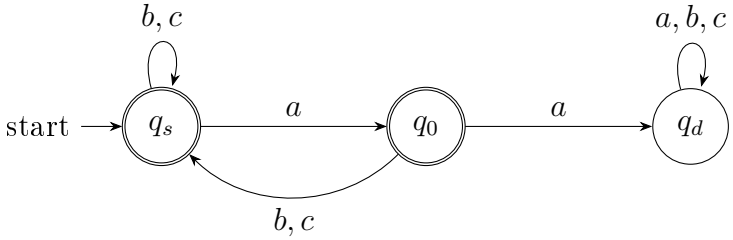


Figure 6

Exercise 18

Description: The set of strings over $\{a, b\}$ that do not begin with the substring aaa.

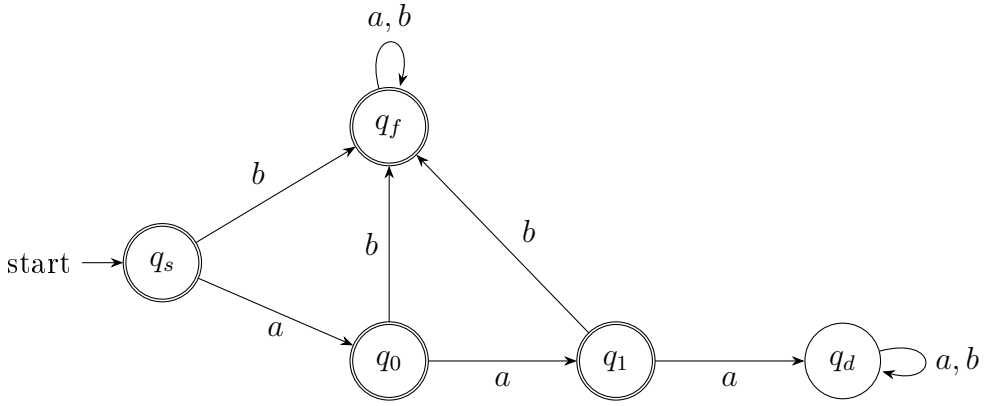


Figure 7

Exercise 19

Description: The set of strings over $\{a, b\}$ that do not contain the substring aaa.

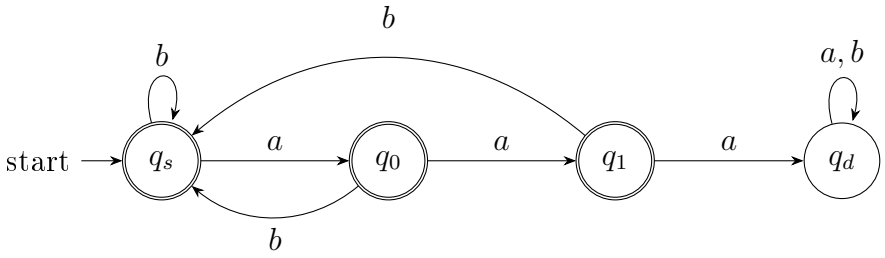


Figure 8

Exercise 20

Description: The set of strings over $\{a, b\}$ that do not contain the substring aba.

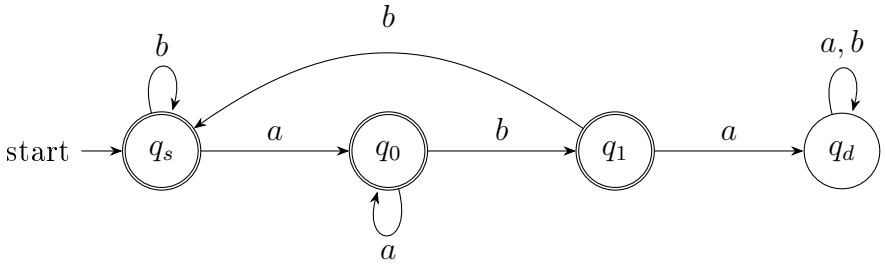


Figure 9

Exercise 21

Description: The set of strings over $\{a, b\}$ in which the substring aa occurs exactly once.

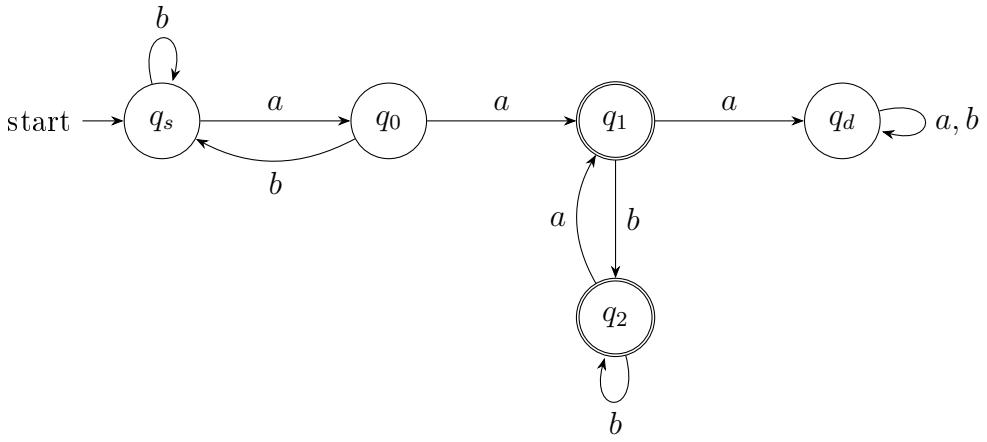


Figure 10

Exercise 22

Description: The set of strings over $\{a, b, c\}$ that begin with a, contain exactly two b's, and end with cc.

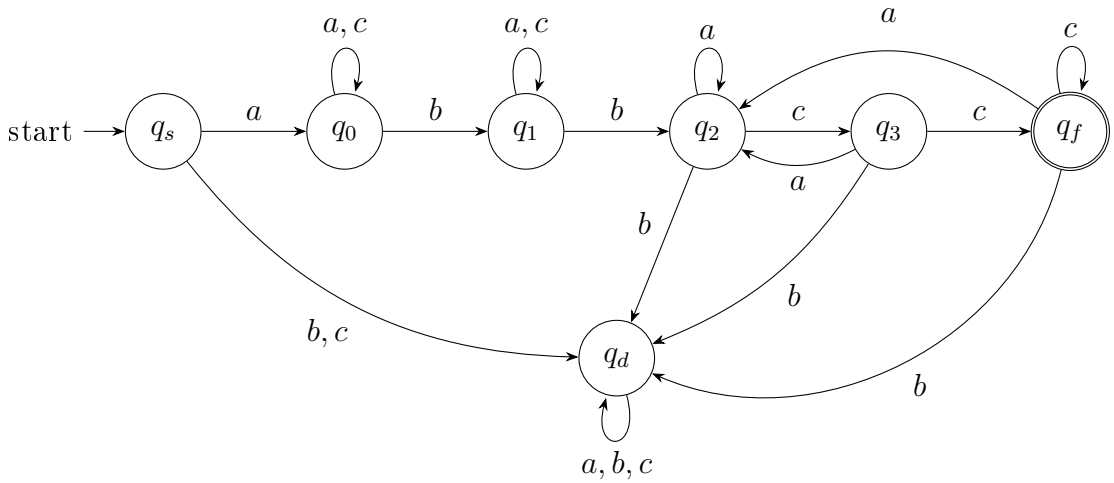


Figure 11

Exercise 23

Description: The set of strings over $\{a, b\}$ that contain the substring ab and the substring ba.

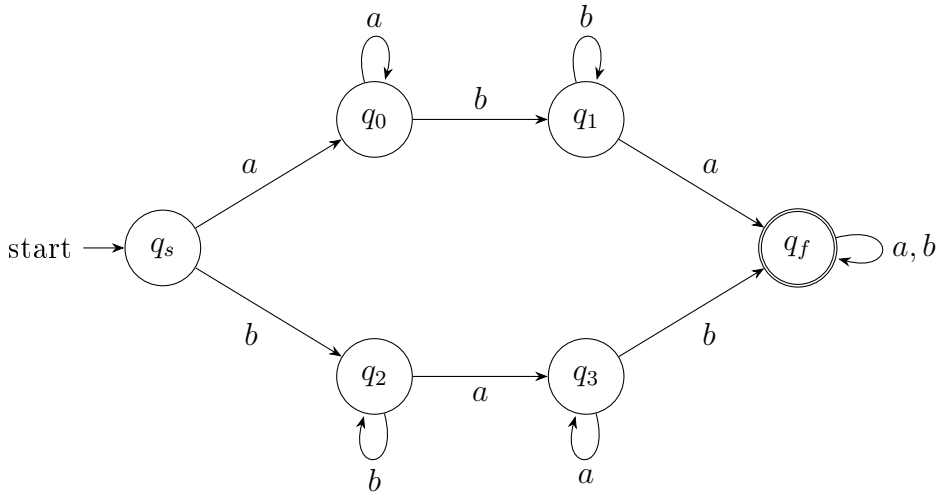


Figure 12

Exercise 24

Description: The set of strings over $\{a, b, c\}$ that contain the substrings aa, bb, and cc.

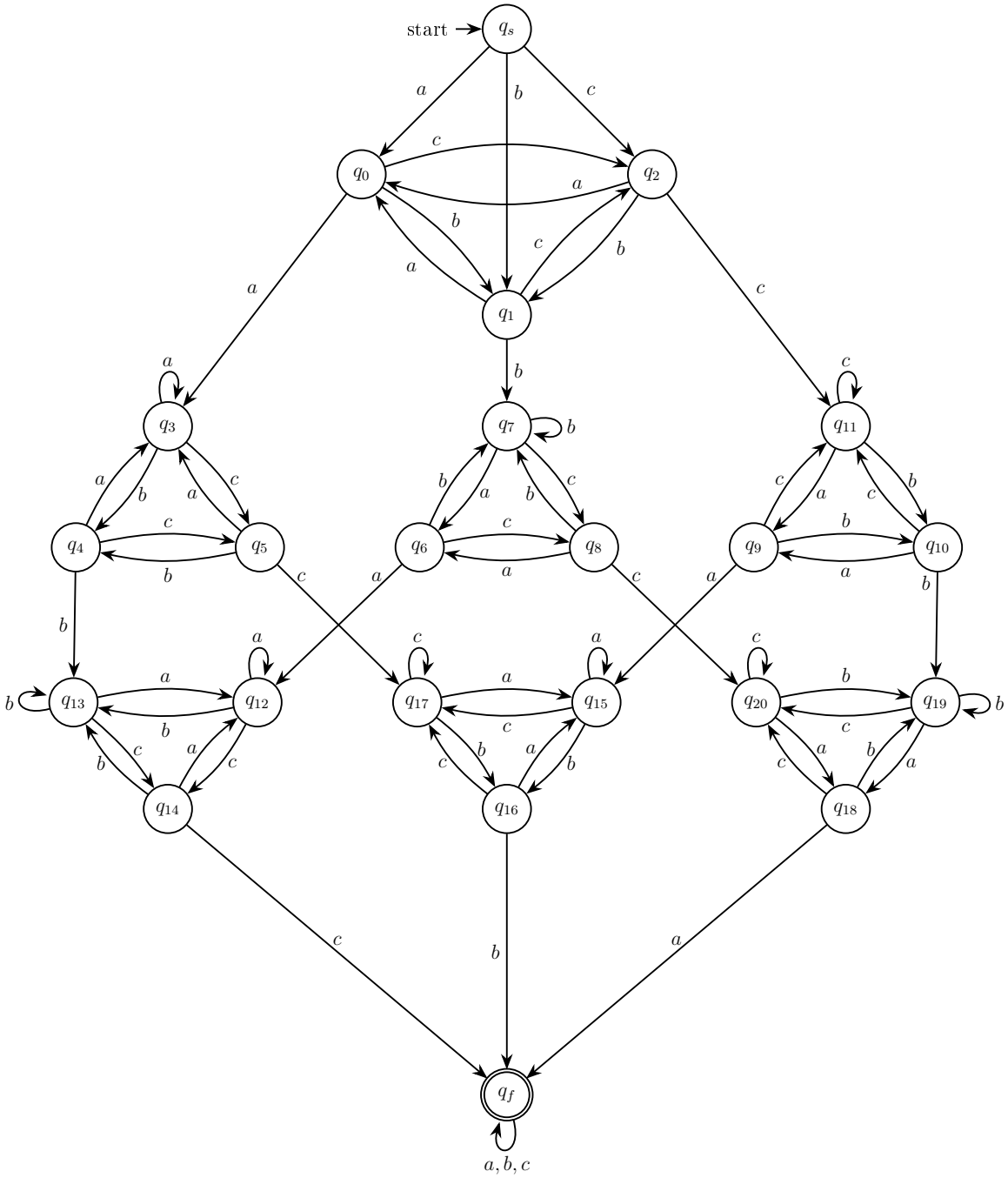


Figure 13

Exercise 25

Description: The set of strings over $\{a, b, c\}$ in which every b is immediately followed by at least one c.

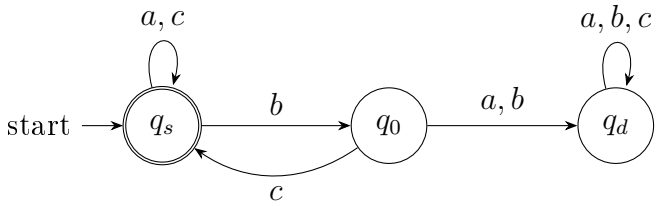


Figure 14

Exercise 26

Description: The set of strings over $\{a, b, c\}$ with length three.

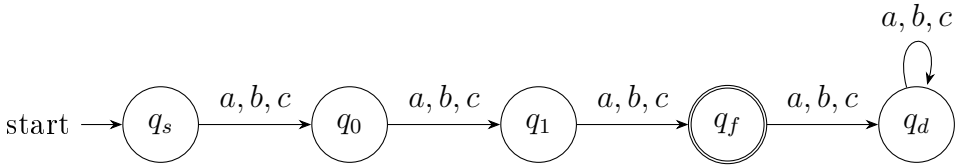


Figure 15

Exercise 27

Description: The set of strings over $\{a, b, c\}$ with length less than three.

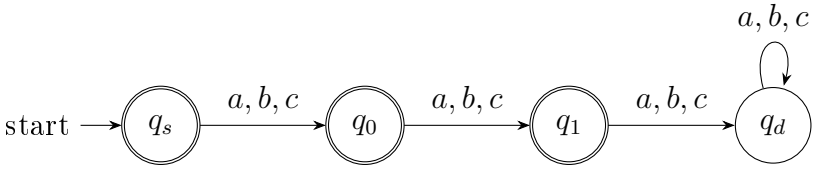


Figure 16

Exercise 28

Description: The set of strings over $\{a, b, c\}$ with length greater than three.

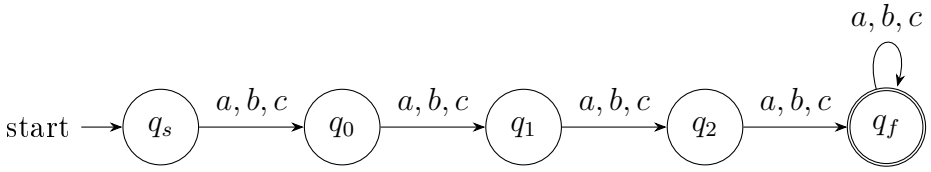


Figure 17

Exercise 29

Description: The set of strings over $\{a, b\}$ in which the number of a's is divisible by three.

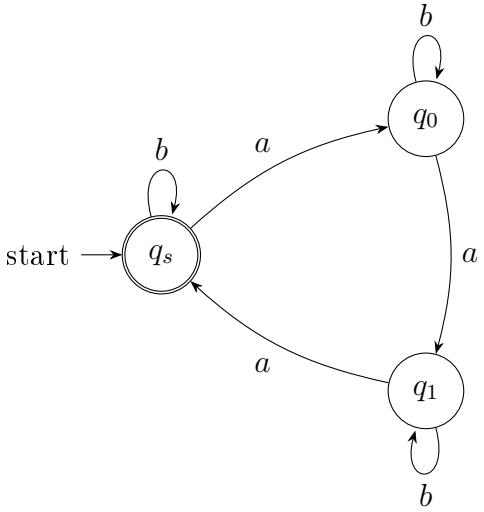


Figure 18

Exercise 30

Description: The set of strings over $\{a, b, c\}$ in which the total number of b's and c's is three.

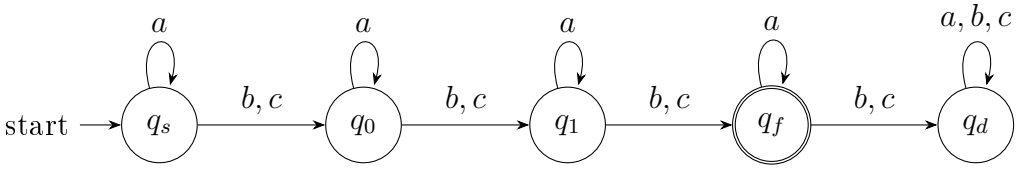


Figure 19

Exercise 31

Description: The set of strings over $\{a, b\}$ in which every a is either immediately preceded or immediately followed by b, for example, $baab$, aba, and b.

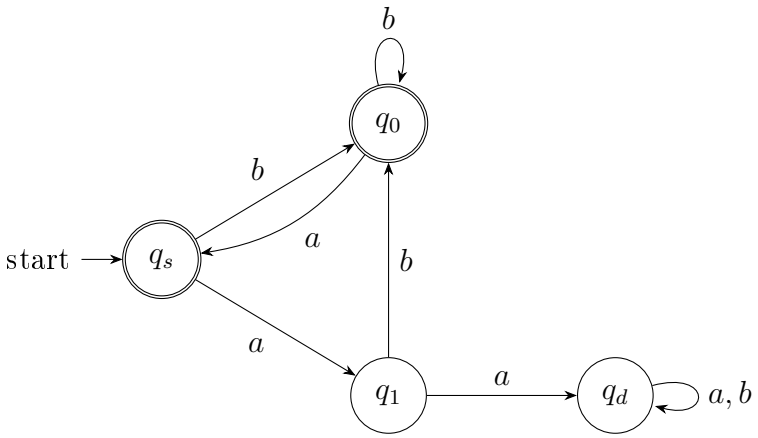


Figure 20

Exercise 32

Description: The set of strings of odd length over $\{a, b\}$ that contain the substring bb.

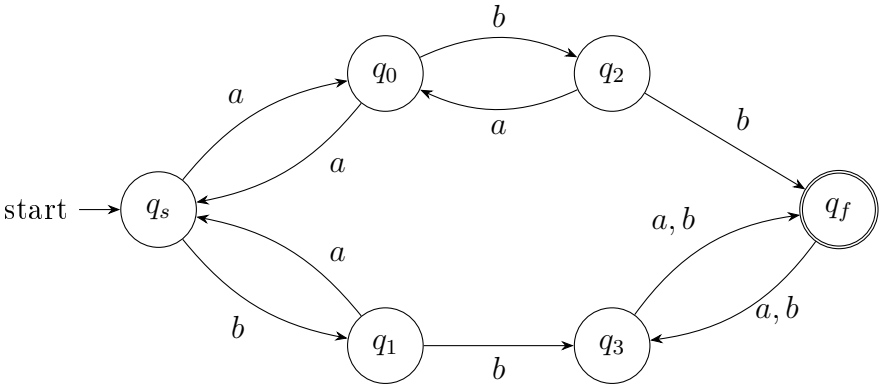


Figure 21

Exercise 33

Description: The set of strings of even length over $\{a, b, c\}$ that contain exactly one a.

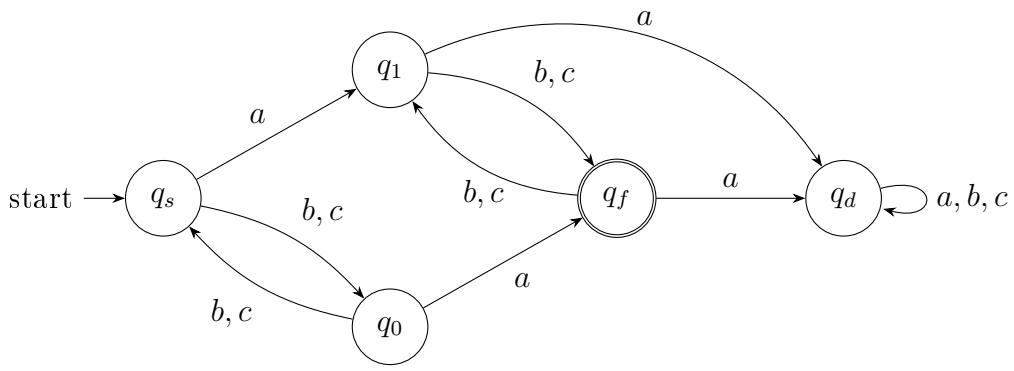


Figure 22

Exercise 34

Description: The set of strings over $\{a, b, c\}$ with an odd number of occurrences of the substring ab.

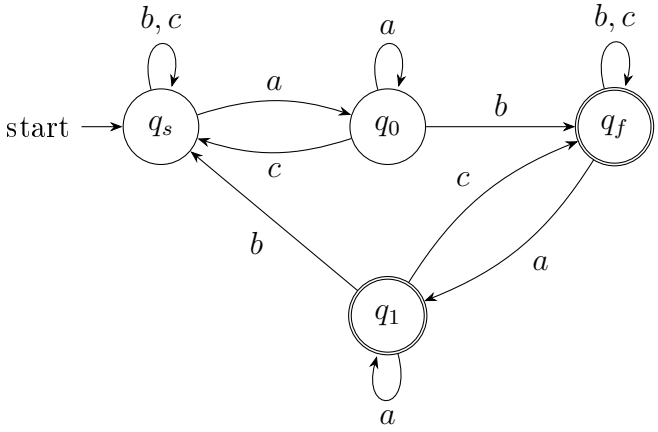


Figure 23

Exercise 35

Description: The set of strings of odd length over $\{a, b\}$ that contain exactly two b's.

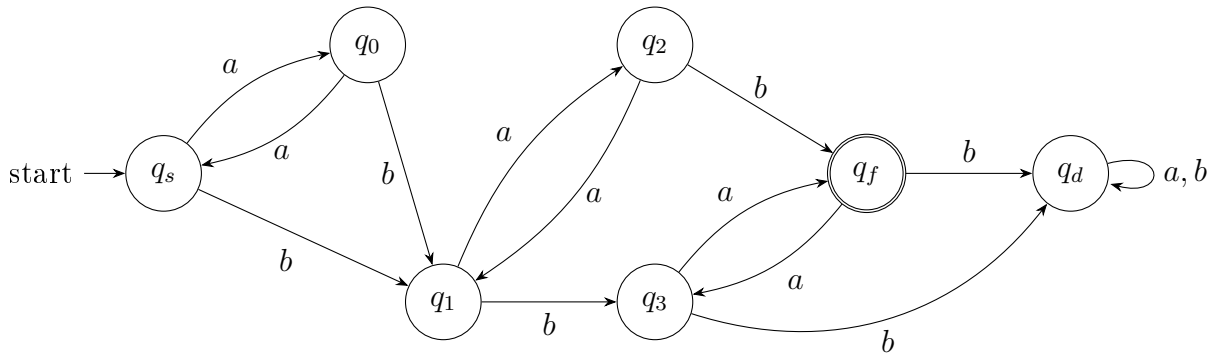


Figure 24

Exercise 36

Description: The set of strings over $\{a, b\}$ with an even number of a's or an odd number of b's.

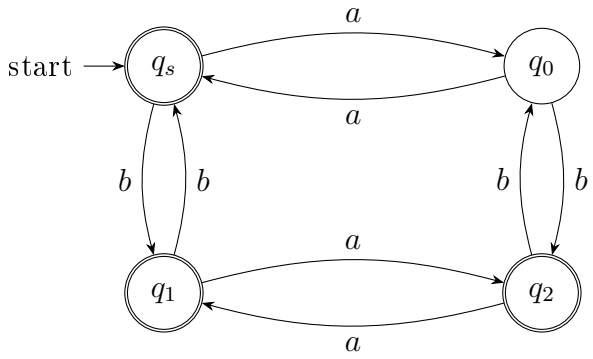


Figure 25